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Does multi-scale GARCH information enhance volatility prediction?

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ABSTRACT

The GARCH-LSTM model is widely used for volatility prediction. However, it fails to account for feature dependencies across different frequencies, and the lack of an attention mechanism limits its predictive accuracy. To address these issues, this paper introduces the GENSHIN framework, which integrates GARCH information with a multi-scale graph neural network to model volatility across multiple frequency bands with MixHop and multi-head attention mechanism. Empirical studies on the volatility of major Chinese stock indices demonstrate that GENSHIN outperforms other deep learning models, such as TimesNet and LSTM. These results highlight the effectiveness of multi-scale information in improving volatility forecasting.

1. Introduction

Prediction of asset volatility is essential for portfolio management and strategic decision-making in the ever-fluctuating financial landscape. Empirical studies suggest that the volatility of financial assets exhibits highly complex behavior including, for example, clustering (Mandelbrot, 1967), asymmetric volatility response (Black, 1976), and leptokurtosis (Carnero et al., 2004), which reflect the long-term autocorrelations and non-linear sequential dependencies of volatility series. In response to this feature of volatility data, Engle (1982) and Bollerslev (1986) introduced Generalized Autoregressive Conditional Heteroscedasticity (GARCH) model to capture some of the intertemporal volatility dynamics, which has become industrial standards. GARCH models have been effective for in-sample stock market volatility estimation but fall short in out-of-sample forecasting. While they capture the clustering of asset price volatility, their predictive accuracy for future volatility is suboptimal, as noted by various studies. Nelson (1992) theoretically demonstrated GARCH's strong in-sample performance but poor out-of-sample predictions. Celik and Ergin (2014) found the HAR-RV model more accurate for Turkish stock market volatility forecasts. However, HAR models also have limitations, such as parameter sensitivity, nonlinearity oversight, frequency dependency, poor long-term forecasting, and overfitting risks. Thus, relying solely on statistical models may not fully meet the needs for predicting stock price volatility.

The need for accurate out-of-sample predictions has led to the use of complex, non-linear machine learning models. For example, Shin and Shin (2022) used LSTM networks to forecast U.S. stock index volatility, incorporating internet search volumes and implied volatility, outperforming traditional models like VAR, VEC, and VECHAR. Similarly, Petrozziello et al. (2022) showed LSTM significantly outperforms R-GARCH and GJR-MEM models, especially during high volatility periods. Wang et al. (2023) found LSTM and Random Forest outperform GARCH in predicting cryptocurrency volatility. Research has also focused on integrating economic variables into deep learning models. Rayadurgam and Mangalagiri (2023) showed combining GARCH with LSTM and ANN enhances

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predictive accuracy in financial contagion forecasting. Building on these, Nguyen et al. (2024) introduced the RECH-X model, a deep learning framework for volatility forecasting with exogenous factors.

However, the GARCH-LSTM model has some inherent limitations. First, it struggles to capture both short- and long-term dependencies in the feature series (Cai et al., 2024). Additionally, GARCH models tend to produce highly collinear features, and the absence of attention mechanisms in LSTM fails to address these interdependencies (Vaswani et al., 2017). To address these issues, this study suggests a GENSHIN (GARCH Estimation based Network for Stock Index Heteroskedasticity INformation) framework. Features are produced by a GARCH-VMD-PLS algorithm, and then the multi-scale graph network is introduced, which applies fast Fourier transformation to capture the fluctuation information across different frequency bands for each feature dimension, and incorporates MixHop (Abu-El-Haija et al., 2019) and multi-head attention mechanisms to detect inner correlations between feature series.

The remainder of this paper proceeds as follows. Section 2 describes the workflow and methodologies of GENSHIN framework. Section 3 is about the experiment setting and result analysis. We conclude the main findings of this paper in Section 4.

2. Methodology

2.1. Model framework

Fig. 1 shows the framework of the proposed GENSHIN model: GARCH Estimation based Network for Stock Index Heteroskedasticity INformation. The detailed modeling procedure is as follows.

(1) GARCH-Extraction for feature mining: GARCH models with different specifications can capture various data characteristics of asset volatility. To fully extract these feature information, we employ standard GARCH models, EGARCH models, GJR-GARCH models, apARCH models, and csGARCH models with lag structures of (1,1), (1,3), (2,1), and (5,5), and error distributions of standard normal, student's t, and generalized error distribution (GED). In total, we utilize 60 GARCH models for volatility modeling, using the GARCH-fitted volatility as a preliminary feature extraction step. The selection of lags (p, q) specifications was guided by three primary principles: (1) adherence to established econometric conventions, (2) comprehensive information capture, and (3) alignment with subsequent feature engineering procedures. The GARCH models with lags $(1, 1)$ represents parsimonious baseline models with remarkable efficacy in capturing volatility clustering. We also consider GARCH models with lags $(1, 3)$ to allow for enhanced modeling of multi-period shock absorption mechanisms in market returns through additional ARCH terms ($q = 3$). Conversely, setting the lags of $(2, 1)$ considers potential higher-order volatility persistence. The lags $(5, 5)$ align with the conventional financial analysis time frame of one trading week.

(2) VMD-PLS feature engineering: Given the nonstationarity and long memory effects of actual volatility data (Mensi et al., 2019) across different time scales, and the high-frequency noise in trading data, we use the variational mode decomposition method (Dragomiretskiy and Zosso, 2013) to decompose GARCH-estimated volatilities into four IMFs each, totaling 240 IMFs. To address overfitting and computational cost when inputting these features into a neural network, we apply partial least squares regression (PLS) to extract six principal components as the model's final inputs.

(3) Model Construction: We adopted MSGNet to efficiently discover and capture the multi-scale inter-series correlations across GARCH features. MSGNet employs several scale-graph blocks, each encompassing three pivotal modules: an FFT module for multi-scale data identification, an adaptive graph convolution module for inter-series correlation learning within a time scale, and a multi-head attention module for intra-series correlation learning.

(4) Model Testing: To illustrate the advantages of the GENSHIN model in volatility forecasting, we introduced six other high-performance time series prediction models and compared their performance with that of the GENSHIN model on the test set. To further demonstrate the model's generalization capability, we consider several popular Chinese stock indices for cross-validation assessments.

2.2. GARCH models

The GARCH model (Bollerslev, 1986) is a statistical model used to analyze time series data, particularly in financial econometrics, where the variance of the error terms is not constant over time.

The GARCH(p, q) model can be expressed as follows:

$$r_t = \mu + \epsilon_t, \quad \epsilon_t = \sigma_t z_t,$$

$$\sigma_t^2 = \omega + \sum_{i=1}^{q} \alpha_i \epsilon_{t-i}^2 + \sum_{j=1}^{p} \beta_j \sigma_{t-j}^2,$$

where r_t is the return at time t, μ is the mean of the process, ϵ_t is the error term at time t, σ_t^2 is the conditional variance at time t, ω, α_i, and β_j are parameters to estimate, z_t is a white noise process with mean zero and variance one.

The error term z_t can follow different distributions, such as standard normal: $z_t \sim N(0, 1)$ and the generalized error distribution: $z_t \sim GED(\mu, \sigma^2, \nu)$, where ν is the shape parameter.

The EGARCH model proposed by Nelson (1991) is defined as,

$$\ln\left(\sigma_t^2\right) = \omega + \sum_{j=1}^{q} \left(\alpha_j z_{t-j} + \gamma_j \left(\left|z_{t-j}\right| - E\left|z_{t-j}\right|\right)\right) + \sum_{j=1}^{p} \beta_j \ln\left(\sigma_{t-j}^2\right).$$

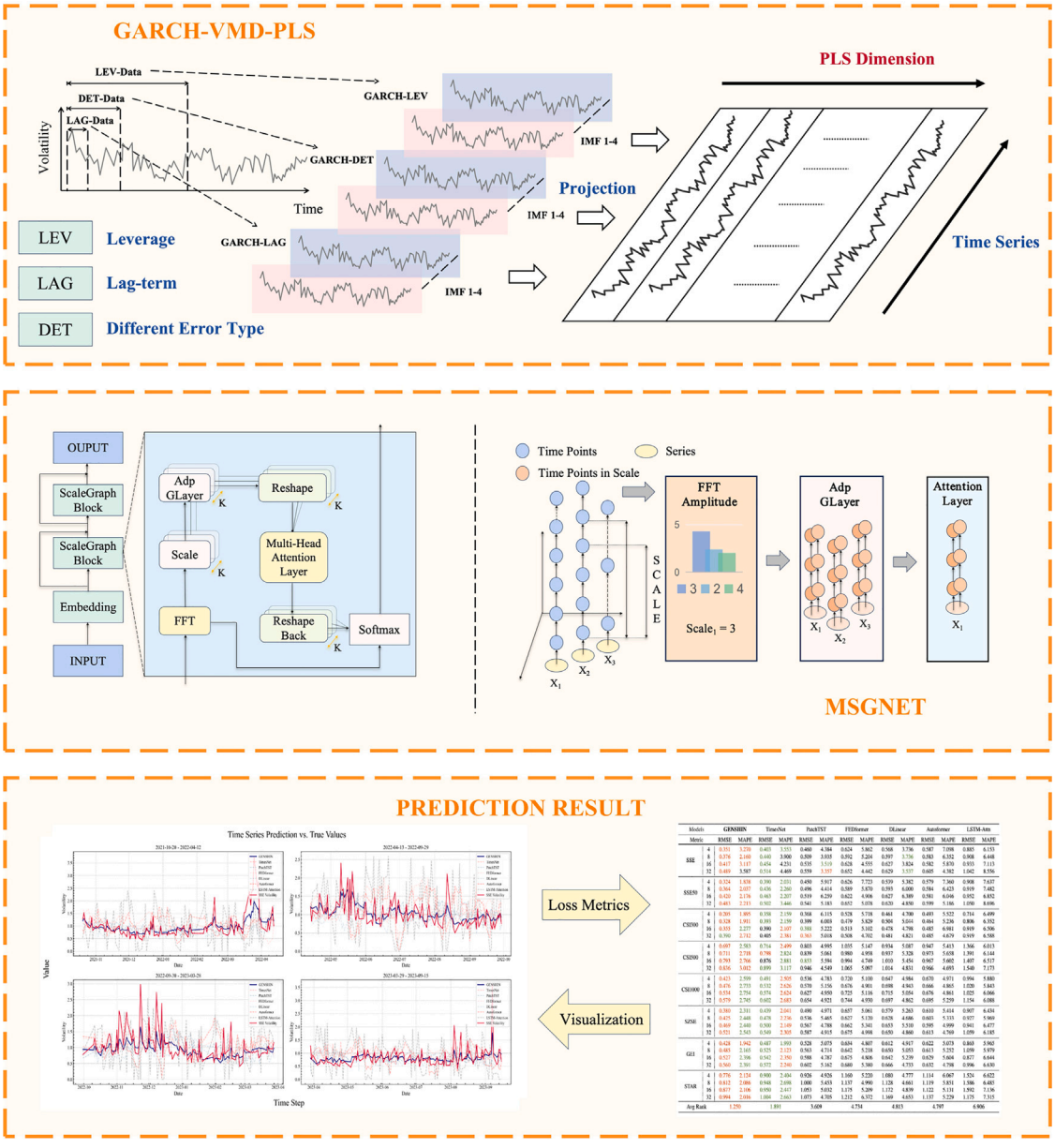


Fig. 1. The work flow of prediction by the GENSHIN framework.

The GJR-GARCH model of Glouston et al. (1993) models positive and negative shocks on the conditional variance asymmetrically via the use of the indicator function I, and can be expressed as:

$$\sigma_t^2 = \omega + \sum_{j=1}^{q} \left(\alpha_j \epsilon_{t-j}^2 + \gamma_j I_{t-j} \epsilon_{t-j}^2 \right) + \sum_{j=1}^{p} \beta_j \sigma_{t-j}^2,$$

where γ_j now represents the 'leverage' term. The indicator function I takes value 1 for $\epsilon \leq 0$ and 0 otherwise.

The component sGARCH model (csGARCH) of Engle and Lee (1999) decomposes the conditional variance into a permanent and transitory component to investigate the long- and short-run movements of volatility affecting securities. Letting q_t represent the permanent component of the conditional variance, the component model can then be written as:

$$\sigma_t^2 = q_t + \sum_{j=1}^{q} \alpha_j (\epsilon_{t-j}^2 - q_{t-j}) + \sum_{j=1}^{p} \beta_j (\sigma_{t-j}^2 - q_{t-j}),$$

$$q_t = \omega + \rho q_{t-1} + \phi(\epsilon_{t-1}^2 - \sigma_{t-1}^2).$$

The asymmetric power ARCH model (apARCH) of Ding et al. (1993) allows for both leverage and the Taylor effect. The apARCH model can be expressed as:

$$\sigma_t^\delta = \omega + \sum_{j=1}^{q} \alpha_j \left(\left| \epsilon_{t-j} \right| - \gamma_j \epsilon_{t-j} \right)^\delta + \sum_{j=1}^{p} \beta_j \sigma_{t-j}^\delta,$$

where $\delta \in \mathbb{R}^+$ is for a Box–Cox transformation of σ_t, and γ_j is the coefficient in the leverage term. Various submodels arise from this model.

2.3. Variational mode decomposition

The variational mode Decomposition (VMD) (Dragomiretskiy and Zosso, 2013) is an advanced time–frequency analysis technique that decomposes a multi-component time series into several single-component amplitude-modulated (AM) and frequency-modulated (FM) signals. The mathematical formulation of VMD is given by the following optimization problem:

$$\min_{\{s_m\},\{\omega_m\}} \left\{ \sum_{m=1}^{M} \left\| \partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) * s(t) \right] e^{-j\omega_m t} \right\|_2^2 \right\} \quad s.t. \sum_m s_m(t) = S(t),$$

where $\delta(t)$ denotes the impulse function, and $*$ represents the convolution operation.

2.4. Partial least squares regression

Partial least squares regression (PLS) is a supervised feature extraction method that utilizes principal component analysis and composite variable extraction to decompose and filter data in a system. It identifies the composite variables that most strongly explain the response variable **Y**, distinguishing between system information and noise, and builds an appropriate model. Based on PLS-driven information feature compression, it compresses the explanatory variable dataset **X** while considering its correlation with the response variable **Y**, resulting in a more informative compression outcome.

2.5. Multi-scale graph neural network

Multi-scale graph neural network (MSGNet) (Cai et al., 2024) is a novel multivariate time series forecasting framework designed to enhance predictive accuracy by capturing inter-series correlations at multiple scales. The core of the model includes the generation of adaptive adjacency matrices, multi-scale graph convolution, and multi-head attention mechanisms, which, when combined, effectively capture and integrate correlations across different time scales, providing high-precision time series forecasting. Next, we briefly introduce some key components of the MSGNet.

The model utilizes the fast Fourier transform (FFT) to detect periodicity, calculated as follows:

$$F = \text{Avg}(\text{Amp}(\text{FFT}(X_{\text{emb}}))),$$

in which $\text{FFT}(\cdot)$ and $\text{Amp}(\cdot)$ denote the FFT and the calculation of amplitude values, respectively. The vector $F \in \mathbb{R}^L$ represents the calculated amplitude of each frequency, which is averaged across d_{model} dimensions by the function $\text{Avg}(\cdot)$. Then, the frequencies are converted into time scales using the following equation:

$$f_1, \ldots, f_k = \text{argTop}_k \left\{ f^* \in \left\{ 1, \ldots, \frac{L}{2} \right\} \right\}(F), \; s_i = \frac{L}{f_i},$$

where argTop_k represents the top k amplitudes of each frequency, and this process aggregates the different scales based on their amplitudes. Then, these tensors are processed through a multi-scale adaptive graph convolution. It projects the tensor corresponding to the ith scale back into a tensor of size N.

$$\boldsymbol{\mathcal{H}}^i = \mathbf{W}^i \boldsymbol{\mathcal{X}}^i.$$

Here, $\boldsymbol{\mathcal{H}}^i \in \mathbb{R}^{N \times s_i \times f_i}$, and $\mathbf{W}^i \in \mathbb{R}^{N \times d_{\text{model}}}$ is a learnable weight matrix, tailored to the ith scale tensor. Then the graph learning process generates two set of trainable parameters, $\mathbf{E}_1^i$ and $\mathbf{E}_2^i \in \mathbb{R}^{N \times h}$. Subsequently, an adaptive adjacency matrix is obtained by multiplying these two parameter matrices following the formula:

$$\mathbf{A}^i = \text{SoftMax}(\text{ReLu}(\mathbf{E}_1^i (\mathbf{E}_2^i)^T)),$$

where the SoftMax function is used to normalize the weights between different nodes.

After obtaining the adjacency matrix $\mathbf{A}^i$ for the ith scale, the MixHop graph convolution method (Abu-El-Haija et al., 2019) is adopted to capture the inter-series correlation. Then a multi-layer perceptron (MLP) is utilized to project the output of MixHop back into a 3D tensor. Moreover, in each time scale, multi-head attention (MHA) is employed to capture the intra-series correlations.

Table 1
Statistical analysis of various stock indices volatility in percent. Count means the number of samples, Mean is the average value, Std is the standard deviation, Skew indicates the skewness of volatility, Kurt refers to the kurtosis of volatility, AR(1) and AR(5) are the first and fifth order autoregressive coefficients respectively, Min and Max are the minimum and maximum values observed, and p_{ADF} is the p value of Augmented Dickey–Fuller test (all p_{ADF} are smaller than the printed values).

Index	Count	Mean	Std	Skew	Kurt	AR(1)	AR(5)	Min	Max	p_{ADF}
SSE	3596	1.121	0.679	3.618	22.987	0.656	0.531	0.276	9.286	0.01
SSE50	3590	1.120	0.682	3.931	27.739	0.647	0.515	0.261	10.164	0.01
CSI300	2984	1.082	0.714	4.135	28.434	0.666	0.545	0.253	9.302	0.01
CSI500	2753	1.231	0.952	4.798	37.346	0.683	0.560	0.295	14.615	0.01
CSI1000	2372	1.214	0.894	3.210	15.588	0.701	0.535	0.346	9.532	0.01
SZSE	3590	1.233	0.706	3.211	18.553	0.661	0.508	0.347	9.180	0.01
GEI	3092	1.535	0.932	4.016	26.373	0.748	0.622	0.457	12.208	0.01
STAR	923	1.350	0.483	2.089	9.083	0.633	0.405	0.535	5.153	0.01

Finally, to proceed to the next layer, MSGNet integrates k different scale tensors $\hat{\boldsymbol{\mathcal{X}}}^1_{\text{out}}, \ldots, \hat{\boldsymbol{\mathcal{X}}}^k_{\text{out}}$ from MHA. The tensor of each scale is first reshaped back to a 2-way matrix $\hat{\mathbf{X}}^i_{\text{out}} \in \mathbb{R}^{d_{\text{model}} \times L}$. Then, MSGNet aggregates the different scales based on their amplitudes:

$$\hat{a}_1, \ldots, \hat{a}_k = \text{SoftMax}(\mathbf{F}_{f_1}, \ldots, \mathbf{F}_{f_k}),$$
$$\hat{\mathbf{X}}_{\text{out}} = \sum_{i=1}^{k} \hat{a}_i \hat{\mathbf{X}}^i_{\text{out}},$$

where $\mathbf{F}_{f_1}, \ldots, \mathbf{F}_{f_k}$ are amplitudes corresponding to each scale, calculated using the FFT. The SoftMax function is then applied to compute the amplitudes $\hat{a}_1, \ldots, \hat{a}_k$.

Ultimately, MSGNet uses the following linear projection formula to transform the output into forecast values:

$$\hat{X}_{t:t+T} = W_s \hat{X}_{\text{out}} W_t + b,$$

where $W_s \in \mathbb{R}^{N \times d_{\text{model}}}$ and $W_t \in \mathbb{R}^{L \times T}$, $b \in \mathbb{R}^T$ are learnable parameters. This design enables MSGNet to effectively capture and integrate correlations across different time scales through adaptive graph convolution and multi-head attention mechanisms, thereby providing high-precision time series forecasting.

3. Experiments

3.1. Experiment setting

3.1.1. Dataset

We compared the performance of GENSHIN with other mainstream deep learning models for eight stock indices as follows: Shanghai Stock Exchange Composite Index (SSE), SSE 50 Index (SSE50), CSI 300 Index (CSI300), CSI 500 Index (CSI500), CSI 1000 Index (CSI1000), Shenzhen Stock Exchange Composite Index (SZSE), Growth Enterprise Index (GEI), STAR Market Index (STAR).

We consider the daily realized volatility (RV) defined as follows:

$$\text{RV}_t = \sqrt{\sum_{i=1}^{M} r_{i,t}^2} \times 100\%,$$

where M is the number of samples per trading day and $r_{i,t}$ is the ith log return on the tth trading day. This definition of volatility is also used by Corsi (2009), Celik and Ergin (2014), Song et al. (2023) and Christensen et al. (2022), etc. We follow Liu et al. (2015) and adopt the frequency of 5-min to compute the RV, and the number of samples involved per day is $M = 48$.

We computed various statistical indicators for the realized volatility of the aforementioned stock indices, as presented in Table 1. The statistical indicators analyzed reveal that the selected index data possess certain commonalities, which can reflect the overall characteristics of market volatility. Additionally, some indices exhibit significant differences in certain indicators compared to others, demonstrating that our chosen sample of data also possesses a degree of diversity.

3.1.2. Baseline models

We consider six alternative time series forecasting methods for comparison, including LSTM-Attn (Vaswani et al., 2017), and Autoformer (Wu et al., 2021), which are based on transformer architectures. In addition, we considered FEDformer (Zhou et al., 2022) based on seasonal-trend decomposition, and DLinear (Zeng et al., 2023) which are linear models. Furthermore, we included PatchTST (Nie et al., 2023), which segments time series into subseries-level patches. Lastly, we considered TimesNet (Wu et al., 2023), which is based on periodic decomposition and currently holds the state-of-the-art performance in time series forecasting. To provide a fair comparison, we also use the GARCH-VMD-PLS procedures for feature engineering to enhance these baseline models.

Table 2
Forecast results with 96 review window and prediction length $4, 8, 16, 32$ in Root Mean Squared Error (RMSE) and Mean Absolute Percentage Error (MAPE). The best result is represented in red, followed by green.

Models		**GENSHIN**		TimesNet		PatchTST		FEDformer		DLinear		Autoformer		LSTM-Attn	
Metric		RMSE	MAPE	RMSE	MAPE	RMSE	MAPE	RMSE	MAPE	RMSE	MAPE	RMSE	MAPE	RMSE	MAPE
SSE	4	0.351	3.270	0.403	3.553	0.460	4.384	0.624	5.862	0.568	3.736	0.587	7.098	0.885	6.153
	8	0.376	2.160	0.440	3.900	0.509	3.935	0.592	5.204	0.597	3.736	0.583	6.352	0.908	6.448
	16	0.417	3.117	0.454	4.231	0.535	3.519	0.628	4.555	0.627	3.824	0.582	5.870	0.933	7.113
	32	0.489	3.587	0.514	4.469	0.559	3.357	0.652	4.442	0.629	3.537	0.605	4.382	1.042	8.556
SSE50	4	0.324	1.838	0.390	2.031	0.450	5.917	0.626	7.723	0.539	5.382	0.579	7.360	0.908	7.637
	8	0.364	2.037	0.436	2.260	0.496	4.414	0.589	5.870	0.593	6.000	0.584	6.423	0.919	7.482
	16	0.420	2.176	0.463	2.207	0.519	6.259	0.622	4.906	0.627	6.389	0.581	6.046	0.952	6.852
	32	0.483	2.213	0.502	3.446	0.541	5.183	0.652	5.078	0.620	4.850	0.599	5.186	1.050	8.696
CSI300	4	0.205	1.895	0.358	2.159	0.368	6.115	0.528	5.718	0.461	4.700	0.493	5.522	0.714	6.499
	8	0.328	1.911	0.393	2.159	0.399	6.003	0.479	5.829	0.504	5.044	0.464	5.236	0.806	6.352
	16	0.355	2.277	0.390	2.107	0.388	5.222	0.513	5.102	0.478	4.798	0.485	6.981	0.919	6.506
	32	0.390	2.712	0.405	2.381	0.363	5.018	0.508	4.702	0.481	4.821	0.485	4.679	0.919	6.588
CSI500	4	0.697	2.583	0.714	2.499	0.803	4.995	1.035	5.147	0.934	5.087	0.947	5.413	1.366	6.013
	8	0.711	2.718	0.798	2.824	0.839	5.061	0.980	4.958	0.937	5.328	0.973	5.658	1.391	6.144
	16	0.793	2.766	0.876	2.881	0.853	5.594	0.994	4.749	1.010	5.454	0.967	5.602	1.407	6.517
	32	0.836	3.012	0.899	3.117	0.946	4.549	1.065	5.097	1.014	4.831	0.966	4.693	1.540	7.173
CSI1000	4	0.423	2.599	0.491	2.505	0.536	4.783	0.720	5.100	0.647	4.984	0.670	4.971	0.994	5.880
	8	0.476	2.733	0.532	2.626	0.570	5.156	0.676	4.901	0.698	4.943	0.666	4.865	1.020	5.843
	16	0.534	2.754	0.574	2.624	0.627	4.950	0.725	5.116	0.715	5.054	0.676	4.861	1.025	6.066
	32	0.579	2.745	0.602	2.683	0.654	4.921	0.744	4.930	0.697	4.862	0.695	5.259	1.154	6.088
SZSE	4	0.380	2.311	0.439	2.041	0.490	4.971	0.657	5.061	0.579	5.263	0.610	5.414	0.907	6.434
	8	0.425	2.448	0.478	2.236	0.536	5.465	0.627	5.120	0.628	4.686	0.603	5.333	0.927	5.969
	16	0.469	2.440	0.500	2.149	0.567	4.788	0.662	5.341	0.653	5.510	0.595	4.999	0.941	6.477
	32	0.521	2.543	0.549	2.305	0.587	4.915	0.675	4.998	0.650	4.860	0.613	4.769	1.059	6.185
GEI	4	0.428	1.942	0.487	1.993	0.528	5.075	0.634	4.807	0.612	4.917	0.622	5.073	0.863	5.965
	8	0.485	2.165	0.525	2.123	0.563	4.714	0.642	5.218	0.650	5.053	0.613	5.252	1.059	5.979
	16	0.527	2.396	0.542	2.350	0.588	4.787	0.675	4.806	0.642	5.239	0.629	5.604	0.877	6.644
	32	0.560	2.391	0.572	2.240	0.602	5.162	0.680	5.380	0.666	4.733	0.632	4.798	0.996	6.630
STAR	4	0.776	2.124	0.900	2.404	0.926	4.926	1.160	5.220	1.080	4.777	1.114	6.067	1.524	6.622
	8	0.812	2.086	0.948	2.698	1.000	5.453	1.137	4.990	1.128	4.661	1.119	5.851	1.586	6.485
	16	0.877	2.106	0.950	2.447	1.053	5.032	1.175	5.209	1.172	4.839	1.122	5.131	1.592	7.136
	32	0.994	2.016	1.004	2.663	1.073	4.705	1.212	6.372	1.169	4.653	1.137	5.229	1.175	7.315
Avg rank		1.250		1.891		3.609		4.734		4.813		4.797		6.906	

3.1.3. Experimental setup

The experiment was conducted using an Apple Mac M3 PRO 18 GB CPU. We use the mean squared error (MSE) as the training loss function. The review window size of all models was set to $L = 96$ (for fair comparison), and the prediction horizons include $T = 4, 8, 16, 32$. These settings were applied to all models. The initial learning rate was LR = 0.0001, batch size was Batch = 32, and the number of epochs was Epochs = 10, and early termination was used where applicable. $70\%, 10\%, 20\%$ of the data are used as the training, validation, and test data, respectively.

3.2. Results and analysis

Table 2 presents a comprehensive comparison of various forecasting models applied to different datasets, with a focus on their root mean squared error (RMSE) and mean absolute percentage error (MAPE) across prediction horizons of 4, 8, 16, and 32 days. The models evaluated include GENSHIN, TimesNet, PatchTST, FEDformer, DLinear, Autoformer, and LSTM-Attn (LSTM with Attention).

GENSHIN and TimesNet show lower RMSE and MAPE, particularly for SSE50 and STAR volatilities. For instance, GENSHIN has an RMSE of 0.324 and MAPE of 1.838 for SSE50 with a 4-day forecast horizon, while TimesNet has an RMSE of 0.390 and MAPE of 2.031. Both models benefit from frequency decomposition, but GENSHIN outperforms TimesNet by using MixHop and multi-head attention. PatchTST ranks third, possibly due to its sub-series patch approach focusing on local temporal features, though not as effective as FFT for capturing multi-frequency information. FEDformer has higher RMSE and MAPE in most cases, suggesting its seasonal decomposition is less effective for short-term financial forecasting. DLinear's linear structure struggles with financial volatility's non-linearity. Autoformer and LSTM-Attn lag behind, likely due to difficulties in capturing complex patterns and dependencies for accurate financial forecasting.

The elevated RMSEs for CSI500, CSI1000, and STAR compared to other indices is mainly due to the higher volatility of small-cap stocks, affecting index stability. However, the MAPEs for these indices are similar to other stock indices, indicating that the increased

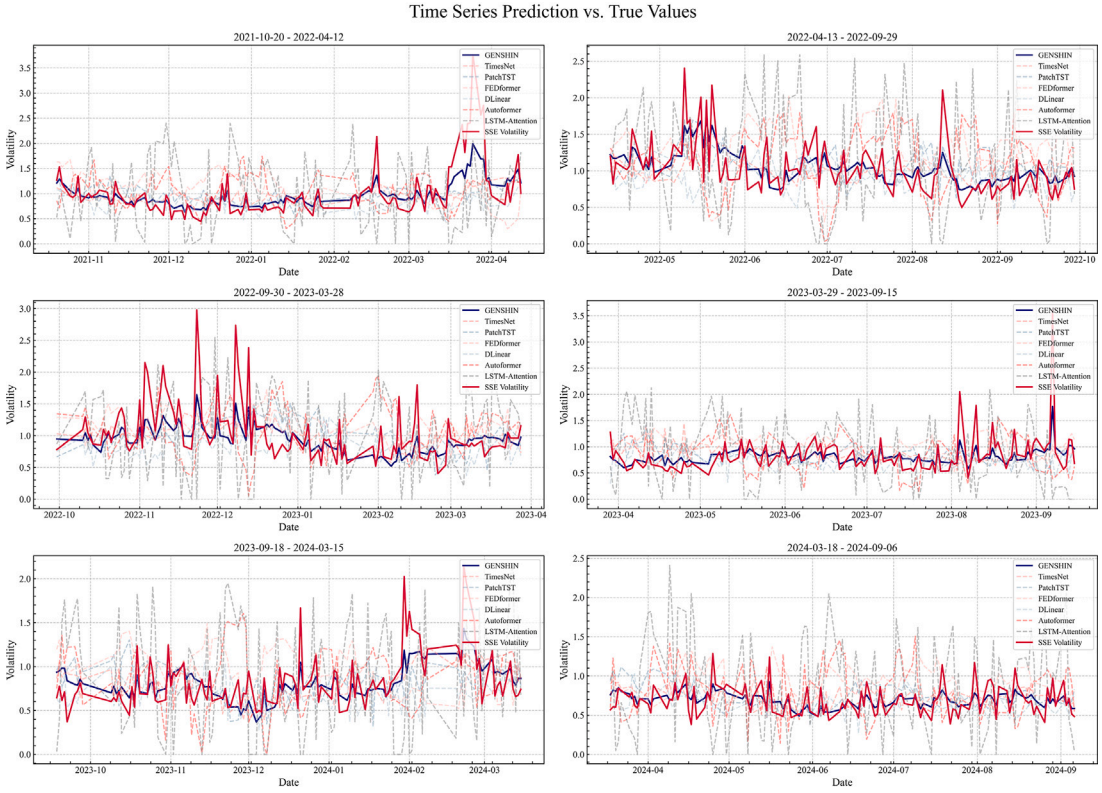


Fig. 2. The comparison of the predictions of 7 deep learning models for SSE volatility with forecasting step 8.

RMSE is likely caused by outliers. The consistently good performance of the GENSHIN model in terms of both RMSE and MAPE across all the indices considered highlights its robustness.

3.3. Visualization

Fig. 2 illustrates a comparative analysis of the performance of seven deep learning models in predicting SSE volatility over various time periods, with a forecasting step of 8. The models compared include GENSHIN, PatchTST, FEDformer, DLinear, Autoformer, LSTM-Attention. The graphs display the predicted values against the true values across six different time frames, ranging from October 2021 to September 2024. Notably, the GENSHIN model consistently demonstrates superior predictive accuracy, closely following the true volatility values across all periods, particularly during volatile periods like 2022-02 and 2023-06. In contrast, other models like TimesNet, PatchTST, FEDformer, and DLinear show larger deviations from the true values, with noticeable differences during peak periods. Autoformer and LSTM-Attention, while better than some others in specific periods, still do not match GENSHIN's prediction accuracy.

4. Conclusion

This paper introduces the GENSHIN (GARCH Estimation based Network for Stock Index Heteroskedasticity INformation) framework, which integrates the GARCH model information with a multi-scale graph neural network to predict the volatility of stock indices. Through empirical studies, we have demonstrated that GENSHIN outperforms other deep learning models, such as TimesNet and LSTM, in forecasting the volatility of major Chinese stock indices, particularly in terms of RMSE and MAPE. For instance, GENSHIN achieved an optimal RMSE of 0.376 and an MAPE of 2.160 for the SSE volatility with a prediction horizon of 8 days, showcasing its superior predictive accuracy compared to other models. For future work, it is interesting to extend the GENSHIN framework for tail risk prediction by either combining it with an ARMA model or directly using tail risk measures as the prediction targets.

CRediT authorship contribution statement

Rentian Yu: Writing – original draft, Software, Methodology, Formal analysis, Conceptualization. **Haotian Xiao:** Writing – original draft, Visualization, Software, Data curation, Conceptualization. **Yukun Zhu:** Writing – review & editing, Writing – original draft, Software. **Gongqiu Zhang:** Writing – review & editing, Supervision, Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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